DESIGN OF CMOS RING OSCILLATOR

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Abstract

Ring oscillators are widely used on chip as compact, self-starting sources of timing signals, since they need no input signal and are built only from inverters. This report presents the design of a 3-stage CMOS ring oscillator in the GPDK 90 nm technology using Cadence Virtuoso, with a target oscillation frequency of $f = 15$ GHz.

VLSI design flow: The design follows a full-custom ASIC flow:
system specification → circuit design → schematic entry → simulation → layout → physical verification → parasitic extraction → post-layout simulation.

Methodology: The transistor widths, PMOS-to-NMOS ratio and supply voltage were selected from parametric sweeps, and the oscillation frequency was first estimated from the measured propagation delay of a single inverter.

Pre-layout: The simulation gives an oscillation frequency of 14.72 GHz with an average power consumption of 182.5 μW. The inverter has a peak gain of about 6, giving a loop gain of about 216 at DC and about 27 at the oscillation frequency, and the stage delays supply the remaining 60° per stage, so the Barkhausen criterion is satisfied.

Post-layout: The layout passes DRC and LVS without errors. After parasitic extraction, the frequency falls to 10.87 GHz at $V_{dd} = 1.2$ V, a reduction of about 26% from the pre-layout value, which is attributed to the wiring capacitance added by the layout. The average power stays at about 180 μW, so the energy per cycle rises from 12.4 fJ to 16.6 fJ.

PVT analysis: At the rated 1.2 V supply, the post-layout oscillator starts in all five process corners, with the frequency ranging from 7.75 GHz (SS) to 13.84 GHz (FF). It keeps oscillating down to a 0.45 V supply (1.6 GHz), with a slope of about 9.3 GHz/V near the rated supply, and its frequency falls by about 27% from 0°C to 100°C.

Conclusion: The project covered the complete full-custom design flow, including process-corner and temperature analysis. The oscillator reaches 14.72 GHz at the schematic level, within about 2% of the 15 GHz target, but falls to 10.87 GHz after layout at 1.2 V because of wiring capacitance. Layout effects should therefore be considered early, and the layout or device sizes re-tuned to recover the lost frequency within the rated supply.

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1 Ring Oscillator

Oscillators are essential components in modern VLSI systems, generating the timing signals required for synchronization, control and data processing [1]. Among the various oscillator architectures, the ring oscillator (RO) stands out due to its simplicity and ease of integration within CMOS technologies [1, 2]. To understand how it works, we first look at its basic building block, the CMOS inverter.

Circuit Diagram

An RO is a chain of several cascaded delay stages in a loop with a unity feedback factor [2]. In a single-ended RO (SERO), an odd number of single-ended inverters are used (Fig. 1.1). The 3-stage ring oscillator consists of three CMOS inverter stages connected in a closed-loop configuration. The output of the final inverter is fed back to the input of the first, creating a regenerative delay loop that sustains continuous oscillation [1].

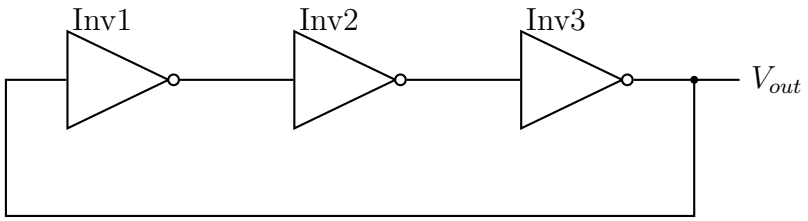


Figure 1.1: Block diagram of the 3-stage ring oscillator.

Fig. 1.1 shows the block diagram, and the transistor-level schematic is given in Fig. 1.2. This simple structure provides high-speed operation and minimal area usage, making it suitable for on-chip clock generation in low-complexity digital systems [1].

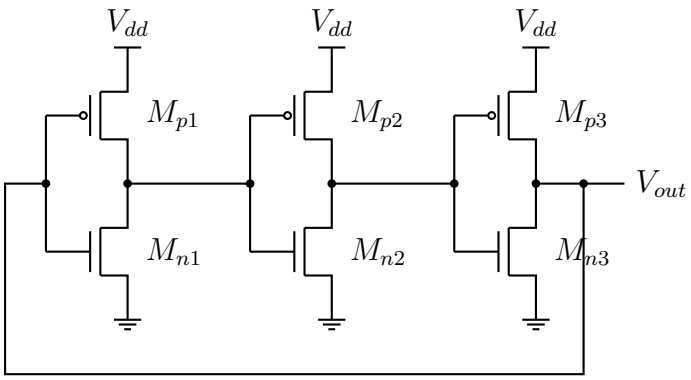


Figure 1.2: Transistor-level schematic of the 3-stage ring oscillator.

1.1 Oscillation Frequency

The common method to calculate the oscillation frequency of an RO is based on the fact that the period of the output signal is twice the sum of the delays of the stages [2], i.e.

$$f = \frac{1}{2Nt_p} = \frac{1}{N\left(t_{pHL} + t_{pLH}\right)} \tag{1.1}$$

where N is the number of stages, t_p is the propagation delay of each inverter (denoted t_d in [2]), t_{pHL} is the high-to-low propagation delay and t_{pLH} is the low-to-high propagation delay. According to Eq. (1.1), to find the oscillation frequency, t_{pHL} and t_{pLH} of each stage should be determined. This is a non-trivial task since there are some parasitics and non-linearities in the delay stages [2].

The oscillation frequency therefore depends on the number of stages (N) and the propagation delay of the inverters. Fewer stages lead to higher oscillation frequencies but may compromise stability and signal quality, whereas more stages provide better waveform shaping and noise tolerance at the expense of reduced speed [1].

1.2 Barkhausen Criterion

For the loop to start and sustain oscillation, the Barkhausen criteria must be satisfied:

1. The loop gain must be at least unity, $|A\beta| \geq 1$ (greater than one at start-up so that oscillations build up, settling to exactly one in steady state).

2. The total phase shift around the loop must be an integer multiple of 360° (i.e. 0° or 360°), so that the feedback is positive.

In a ring oscillator, each inverter provides a 180° inversion. With an odd number of inverters, the loop has a net inversion of 180°. The remaining 180° must come from the propagation delay of the stages, so that the total phase shift is 360° [1]. Each of the N stages therefore contributes a delay-induced phase shift of

$$\Delta\phi = \frac{180^\circ}{N} \tag{1.2}$$

For the 3-stage ring oscillator used in this project ($N = 3$), each stage contributes

$$\Delta\phi = \frac{180^\circ}{3} = 60^\circ \tag{1.3}$$

An even number of inverters would give no net inversion, so the loop would have two stable states and latch instead of oscillating; this is why an odd N is used [2].

2 Design of Ring Oscillator

2.1 Design Parameters and Specification

The oscillation frequency of a ring oscillator depends on the number of stages (N) and the propagation delay of the inverters, as given by Eq. (1.1) [2]. In a SERO, the designer can change the transistor sizes, the supply voltage and the number of stages [2]. The PMOS width is chosen as a ratio of the NMOS width, $W_p = R\,W_n$, and the channel lengths are equal, $L_n = L_p = L$ [2].Over a wide range, W_n has a negligible effect on the frequency: a larger W_n strengthens the transistors but also loads the next stage, and the two effects cancel [2]. The general form of the frequency is therefore [2]

$$f = f(R,\, V_{dd},\, L,\, N), \tag{2.1}$$

where $R = W_p/W_n$ is the PMOS-to-NMOS width ratio, V_{dd} is the supply voltage, L is the channel length and N is the number of stages [2]. For a single-ended ring oscillator, the frequency can be estimated as [2]

$$\begin{aligned} f(R, V_{dd}, L, N) &= \frac{1}{N}\, y(R)\, x(V_{dd})\, g(L), \\ y(R) &= a_1 R^3 + b_1 R^2 + c_1 R + d_1, \\ x(V_{dd}) &= k_1 \ln(V_{dd}) + k_2, \\ g(L) &= m_1 L^{-m_2}, \end{aligned} \tag{2.2}$$

where the coefficients are obtained from simulation results by regression [2].

The parameters of this project are listed in Table 2.1. N and L are fixed by the design choice and the technology, while W_n, R and V_{dd} are selected from the sweeps.

Table 2.1: Design parameters of the 3-stage ring oscillator.

Parameter	**Value**	**Reason**
N	3	The number of stages must be odd for oscillation [2].
L	100 nm	Channel length of all transistors, chosen according to the GPDK 90 nm technology.

Since N and L are fixed, the frequency of this design depends on

$$f = f(W_n,\, R,\, V_{dd}). \tag{2.3}$$

2.2 Design Calculations

2.2.1 CMOS Inverter

A CMOS inverter consists of one PMOS and one NMOS transistor connected in the standard configuration. When the input is low, the PMOS conducts and pulls the output up to V_{dd}; when the input is high, the NMOS conducts and pulls the output down to ground. Typically, the width of the PMOS is chosen to be a ratio of the width of the NMOS, i.e. $W_p = R\,W_n$ [2]. Each inverter introduces a propagation delay, and this delay is what sets the oscillation frequency of the ring [1].

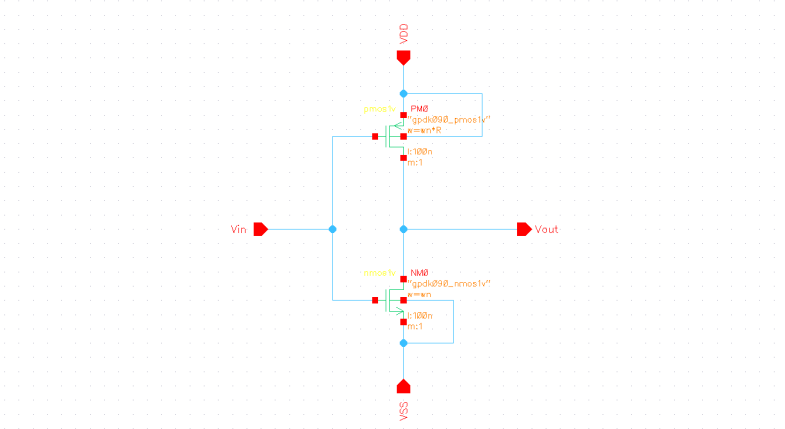


Figure 2.1: CMOS inverter used in each stage (GPDK 90 nm).

2.2.2 3-Stage Ring Oscillator

Three such inverters are connected in a closed loop, with the output of the third fed back to the input of the first, as shown in Fig. 2.2.

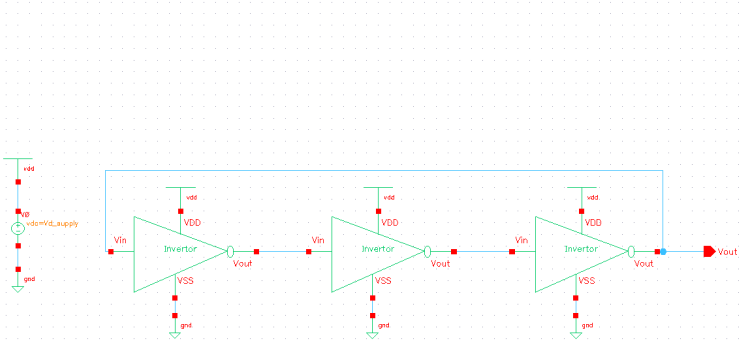


Figure 2.2: 3-stage ring oscillator.

2.2.3 Parameter Selection from Simulation

The parameters W_n, R and V_{dd} are selected from parametric sweeps of the oscillation frequency, varying one parameter at a time with N, L and the others fixed.

NMOS width W_n. The curve in Fig. 2.3 is nearly flat, so W_n has only a small effect on the frequency [2]. The plot gives $W_n = 228$ nm for the target $f = 15$ GHz.

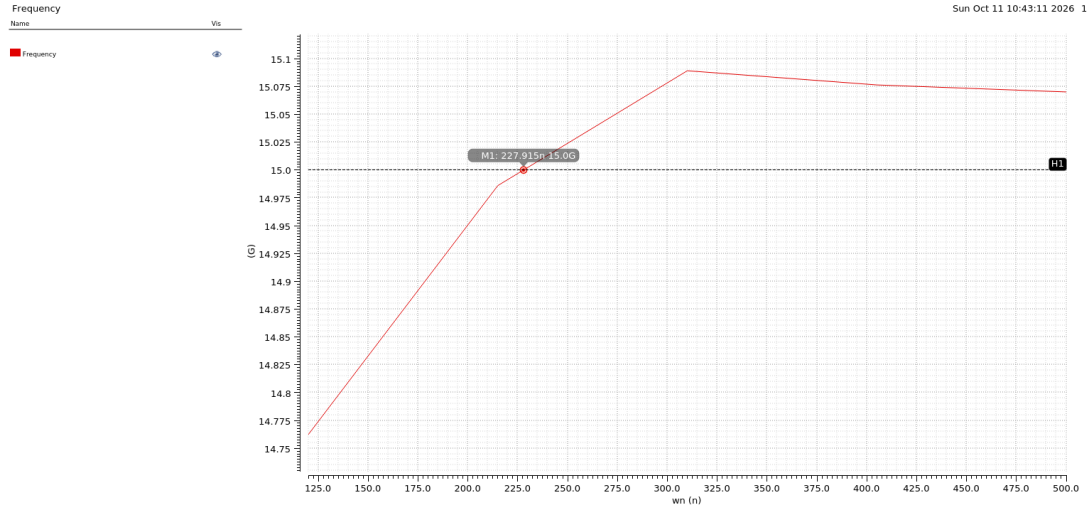


Figure 2.3: Oscillation frequency versus W_n.

PMOS-to-NMOS ratio R. In Fig. 2.4, the frequency rises with R, peaks, and then falls, because a larger PMOS increases the drive current but also loads the previous stage [2]. The maximum of 14.48 GHz occurs at $R = 2.7$, so $W_p = R\,W_n \approx 615.6$ nm.

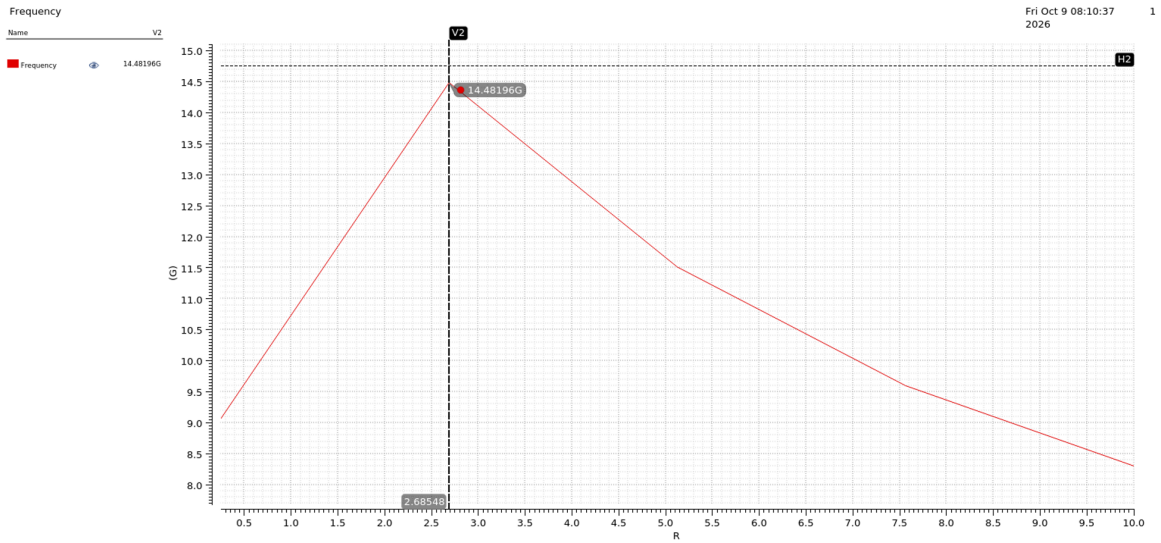


Figure 2.4: Oscillation frequency versus R.

Supply voltage V_{dd}. In Fig. 2.5, the frequency increases with V_{dd} because of the larger gate overdrive [2]. The plot reaches 15 GHz at 1.245 V, which is rounded to the nominal $V_{dd} = 1.2$ V.

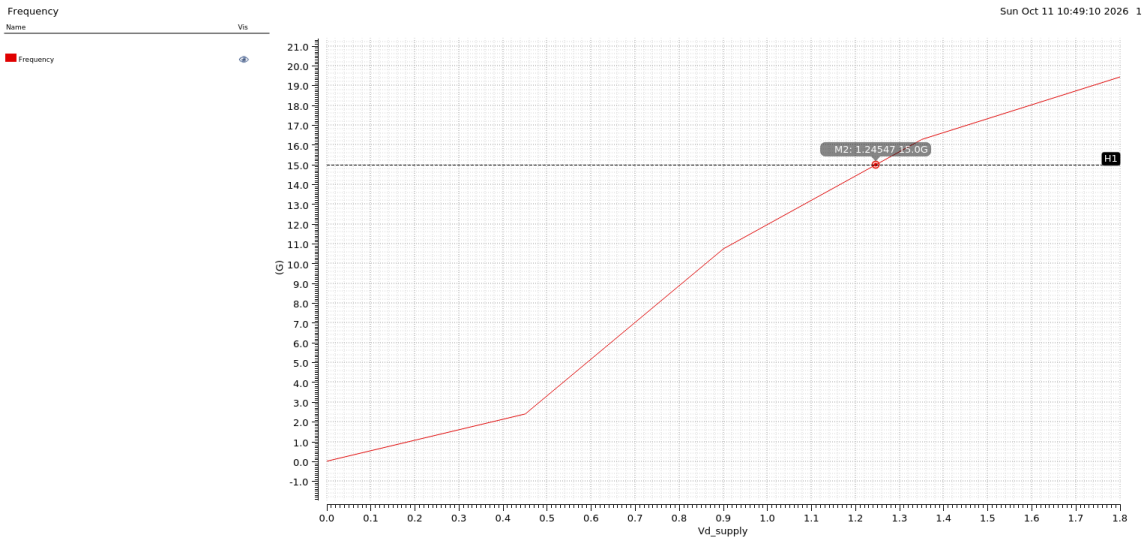


Figure 2.5: Oscillation frequency versus V_{dd}.

2.3 Final Circuit Design

Among these parameters, W_n has a negligible effect on the oscillation frequency, whereas R, V_{dd}, L and N determine it [2]. The supply voltage has the largest effect on the power consumption, and it is therefore the most effective parameter for reducing power [2].

The final design parameters obtained from the above calculations are summarised in Table 2.2.

Table 2.2: Final design parameters of the 3-stage ring oscillator.

S.No	Parameter	Design
1	W_n	228 nm ≈ 230 nm
2	W_p	616 nm ≈ 615 nm
3	L	100 nm
4	V_{dd}	1.2 V
5	N	3

3 Theoretical Calculation

The oscillation frequency of a ring oscillator is set by the propagation delay of its inverters [2, 1]. In this chapter the delays of a single inverter are measured and used to calculate the frequency of the final design, which is then compared with the simulated value. The inverter uses $W_n = 228$ nm, $R = 2.7$, $L = 100$ nm and $V_{dd} = 1.2$ V.

3.1 Inverter Testbench

The inverter is driven by a pulse source swinging from 0 to 1.2 V with a rise time of 1 ns, and it is supplied from $V_{dd} = 1.2$ V, as shown in Fig. 3.1. A transient analysis is run from 0 to 200 ns. The input edge is much slower than the delay being measured, so the delays are taken for a single inverter without the loading of a following stage.

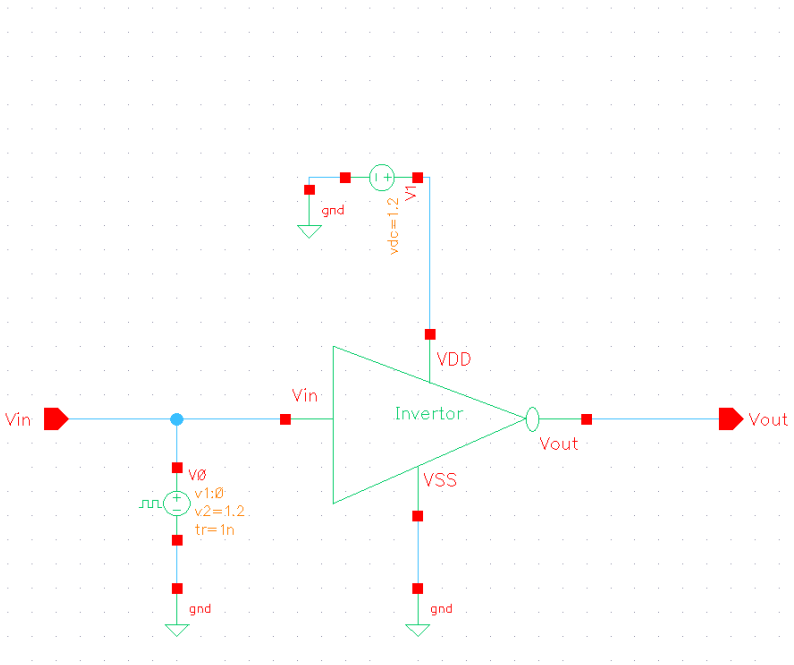


Figure 3.1: Test bench of the CMOS inverter.

3.2 Input and Output Waveforms

The resulting waveforms are shown in Fig. 3.2. The output is the inverted copy of the input. The time from the 50% point of the input to the 50% point of the output gives the high-to-low delay t_{pHL} and the low-to-high delay t_{pLH} [2].

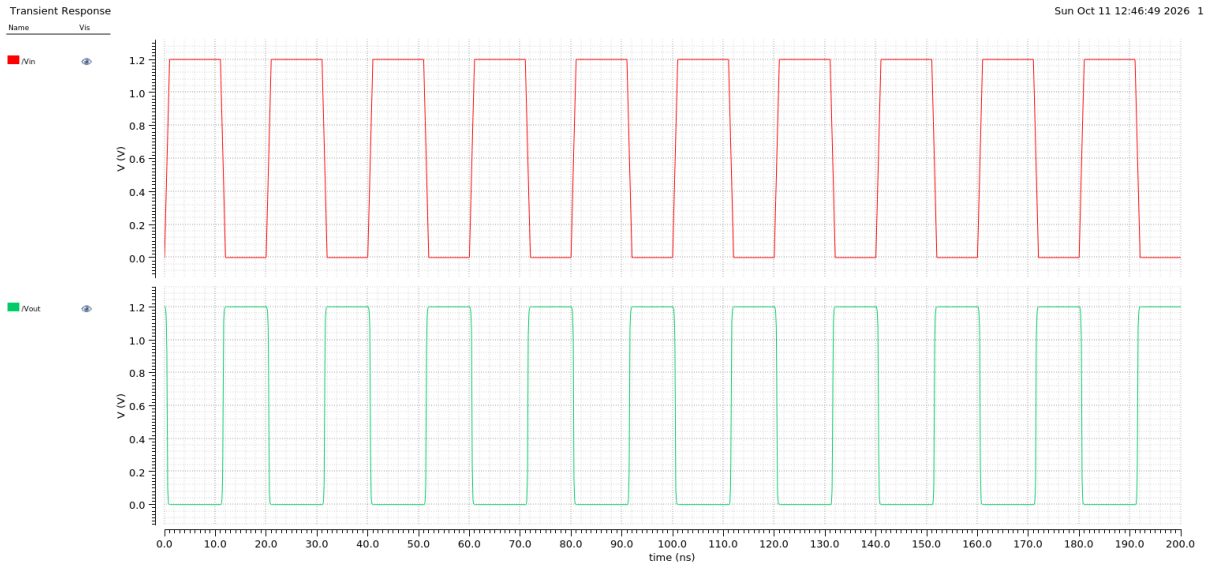


Figure 3.2: Input and output waveforms of the CMOS inverter.

Propagation Delay : The delays are measured with the simulator calculator, as shown in Fig. 3.3: $t_{pHL} = 8.589$ ps and $t_{pLH} = 14.154$ ps. The rising transition is slower, because the PMOS charges the output with a smaller current than the NMOS discharges it, even with $R = 2.7$.

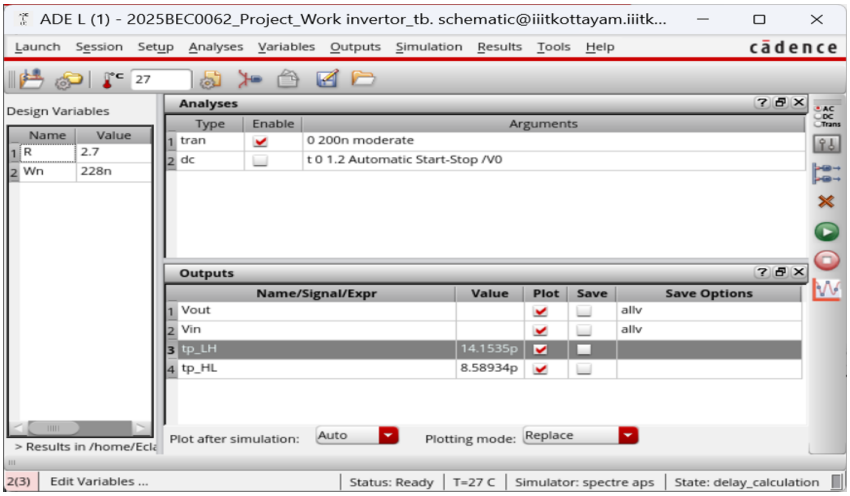


Figure 3.3: Simulation set-up and measured propagation delays.

Loop gain: A DC sweep of the inverter (Fig. 3.4) gives a trip voltage of about 0.60 V and a peak small-signal gain $|A_0| \approx 6$ near the trip point. In the closed ring every node sits at this point, so the DC loop gain is $|A_0|^3 \approx 216 \gg 1$ and any small noise at start-up grows around the loop until the amplitude is limited by the supply and the transistor nonlinearity. At the oscillation frequency each stage must add 60° of delay phase; in a single-pole model this reduces the stage gain to $|A_0| \cos 60° = |A_0|/2 \approx 3$, so the loop gain stays at $(|A_0|/2)^3 \approx 27 > 1$ (single-pole estimate). The three inverting stages give 180°

and the three stage delays give the other 180° (60° per stage), so the total loop phase is 360° and the Barkhausen criterion is satisfied.

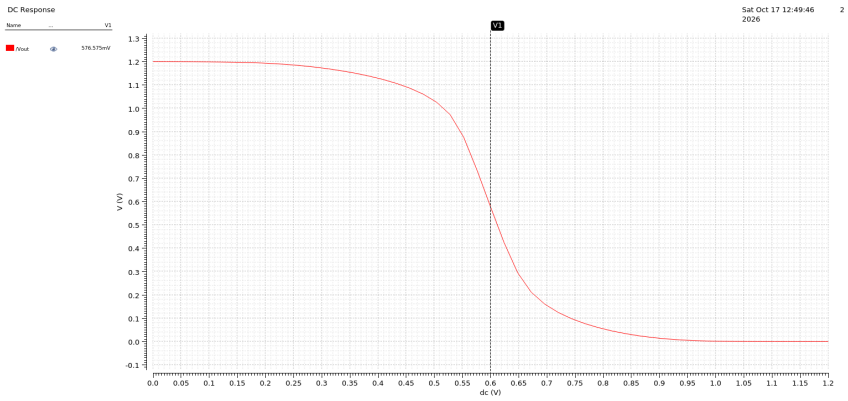


Figure 3.4: DC transfer characteristic of the CMOS inverter

3.3 Result

The rising edge ($t_{pLH} = 14.15$ ps) is about 1.65 times slower than the falling edge ($t_{pHL} = 8.59$ ps), because the PMOS is weaker than the NMOS. $R = 2.7$ was chosen for maximum frequency, not for symmetry; a symmetric inverter would need R between 2 and 4 [2].

Each stage is crossed twice per period, so the oscillation frequency follows from Eq. (1.1):

$$f = \frac{1}{N\,(t_{pHL} + t_{pLH})} \tag{3.1}$$

which is the same as $f = 1/(2\,N\,t_p)$ with the average delay $t_p = (t_{pHL} + t_{pLH})/2$. Substituting the measured delays and $N = 3$ gives the values in Table 3.1.

Table 3.1: Calculated oscillation frequency.

Parameter	**Calculation**	**Unit**
t_{pHL}	8.58934	ps
t_{pLH}	14.1535	ps
$t_p = (t_{pHL} + t_{pLH})/2$	11.37142	ps
N	3	
$f = 1/(2\,N\,t_p)$	14.65662746	GHz

The calculated 14.66 GHz is about 2.3% below the 15 GHz target. The difference is expected, since the delays were measured on an isolated inverter, whereas in the ring each inverter drives the next stage [2].

4 Circuit Simulation and Analysis

The final 3-stage ring oscillator is simulated with $W_n = 230$ nm, $W_p = 615$ nm, $L = 100$ nm and $V_{dd} = 1.2$ V (Table 2.2).

4.1 Schematic and Symbol

Three inverters are connected in a closed loop, with the output of the third stage fed back to the input of the first (Fig. 4.1). This creates a regenerative delay loop that sustains oscillation [1]. **There is no input signal.**

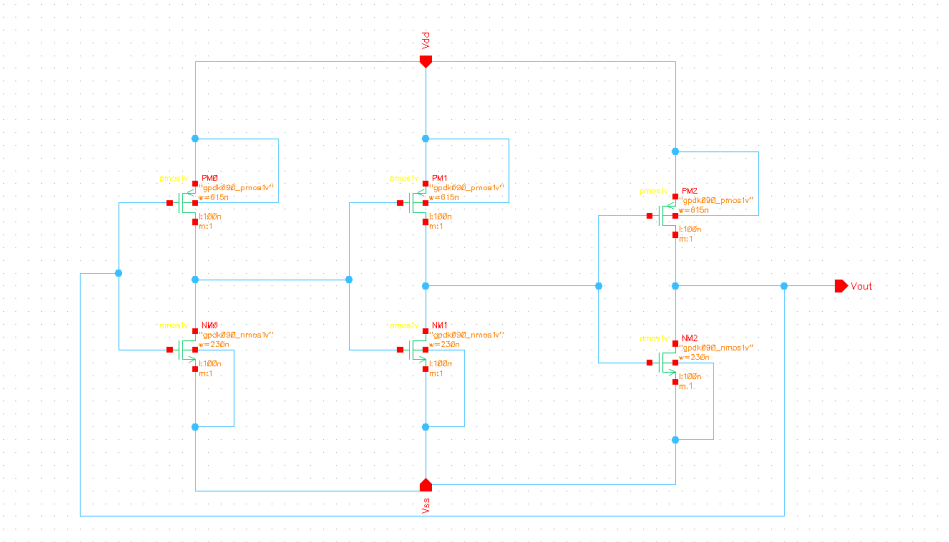


Figure 4.1: Transistor-level schematic of the final 3-stage ring oscillator.

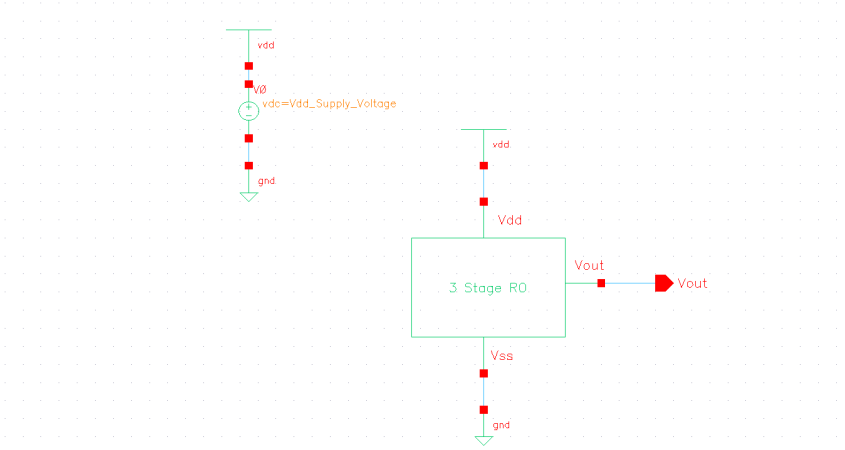


Figure 4.2: Symbol of the 3-stage ring oscillator

The schematic is packaged as a symbol (Fig. 4.2) so it can be reused in a test bench. The symbol has three pins: `Vdd` (supply), `Vss` (ground) and `Vout` (output). It has no input pin, since the oscillator generates its own signal.

In the test bench, a DC source `V0` (`vdc = Vdd_Supply_Voltage`) is connected between `Vdd` and `gnd`.

In an ideal circuit the loop would remain at its unstable equilibrium point, so in the simulation the output is set to $V_{out} = 0$ as an initial condition, which starts the oscillation.

The initial condition is applied in the simulator under *ADE-L Window* → *Simulation* → *Convergence Aids* → *Initial Condition*. In a real circuit, noise starts the oscillation instead, and the amplitude is limited by the supply rails.

4.2 Transient Response

The output of each stage at 27°C is shown in Fig. 4.3. All three waveforms have the same period and each lags the previous stage, which confirms sustained oscillation. The edges are rounded and the waveforms are not ideal square waves.

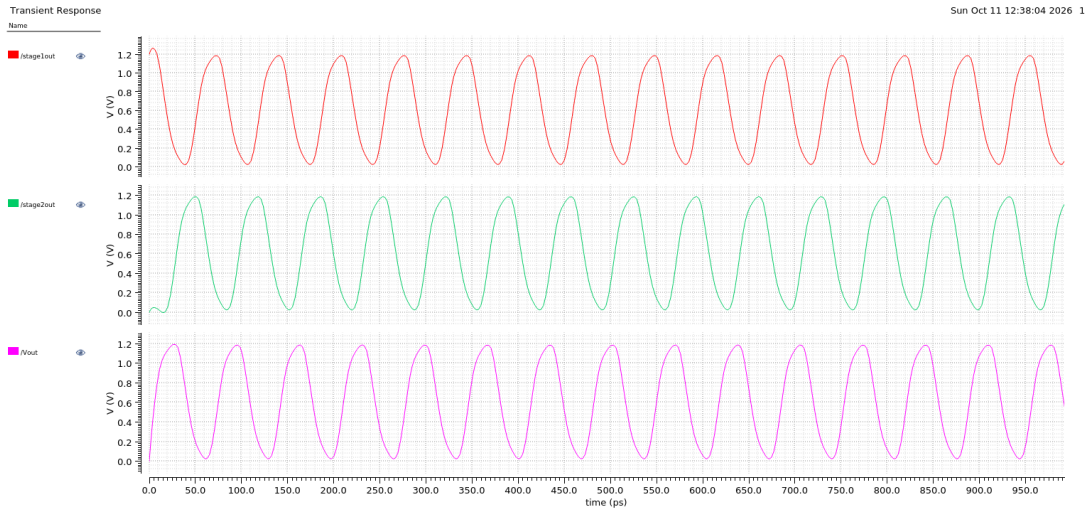


Figure 4.3: Transient response of the output of each stage.

4.3 Frequency of Ring Oscillator

The frequency measured with the simulator calculator (Fig. 4.4) is 14.72 GHz. This is within about 0.4% of the calculated value of 14.66 GHz (Section 3.3) and about 1.9% below the 15 GHz target.

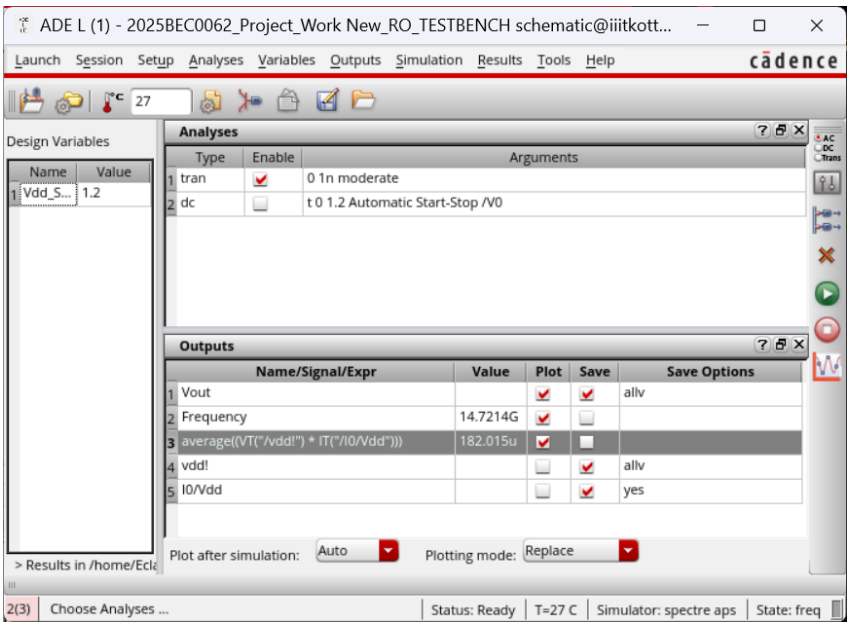


Figure 4.4: Simulation set-up and measured frequency of the ring oscillator.

Fig. 4.5 shows the frequency against time. It rises from 14.46 GHz to 14.69 GHz within about 75 ps as the oscillation builds up, then stays between 14.69 and 14.72 GHz, a variation of only 0.1%.

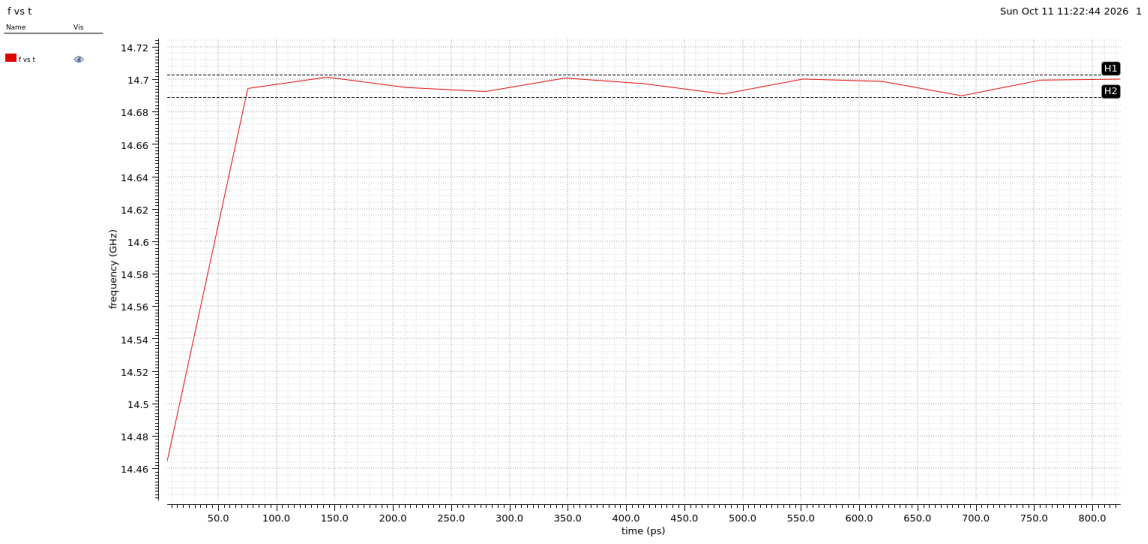


Figure 4.5: Frequency of the ring oscillator versus time.

4.4 Performance Analysis

Power consumption : The instantaneous power (supply voltage times the sum of the three PMOS currents) is shown in Fig. 4.6. After a start-up peak of about 310 μW, it varies periodically between about 165 and 203 μW as the inverters switch.

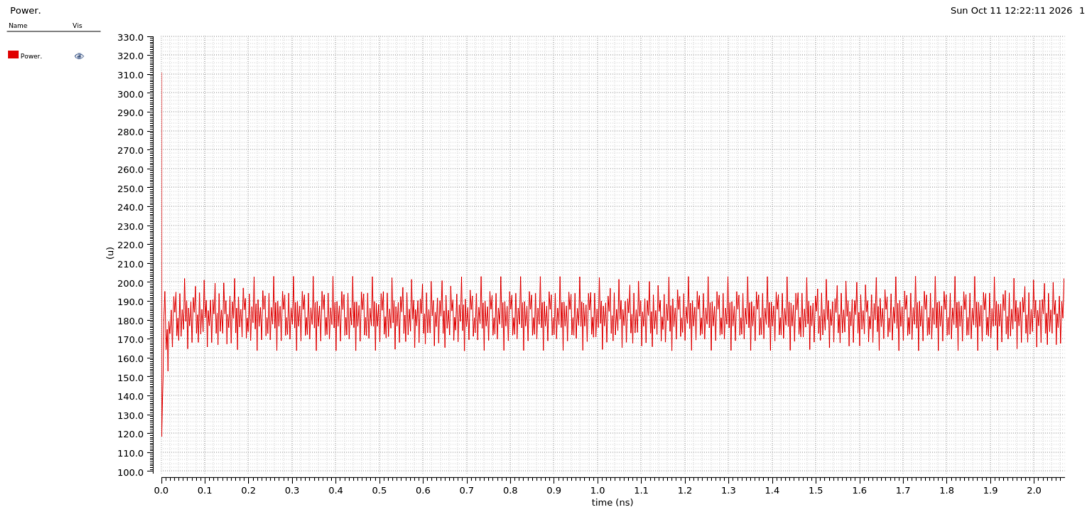


Figure 4.6: Instantaneous power consumption versus time.

The average power (Fig. 4.4) is

$$P_{avg} = 182.5\ \mu\text{W}, \tag{4.1}$$

the energy per cycle is

$$E = \frac{P_{avg}}{f} = \frac{182.5\ \mu\text{W}}{14.72\ \text{GHz}} \approx 12.4\ \text{fJ}. \tag{4.2}$$

V_{dd} has the largest effect on power [2], so lowering the supply is the most effective way to reduce power, at the cost of frequency.

Phase shift : A stage delay t_p corresponds to a phase shift

$$\phi = 360^\circ\, f\, t_p. \tag{4.3}$$

With $t_p = 11.37$ ps and the simulated $f = 14.72$ GHz,

$$\phi = 360^\circ \times 14.72 \times 10^9 \times 11.37 \times 10^{-12} \approx 60.2^\circ, \tag{4.4}$$

which matches $180^\circ/N = 60^\circ$ of Eq. (1.2). The three stages give 180° of delay phase, and the odd number of inverting stages adds another 180°, so the loop phase is 360°, satisfying the Barkhausen criterion [1].

4.5 Results

Table 4.1 summarises the results.

Table 4.1: Simulated results (pre-layout).

Parameter	**Value**
Target frequency	15 GHz
Calculated frequency	14.66 GHz
Simulated frequency	14.72 GHz
Average power	182.5 μW
Energy per cycle	12.4 fJ
Phase shift per stage (calculated)	60°
Phase shift per stage (simulated)	60.2°

These are pre-layout results. Layout parasitics add capacitance and lower the frequency; in [2] the post-layout frequency of a minimum-size oscillator was about 0.7 of the schematic value.

5 Layout Design

5.1 Layout

The layout uses $W_n = 230$ nm, $W_p = 615$ nm and $L = 100$ nm. Three identical inverters are placed side by side with the same orientation, so every stage sees the same wiring and the mismatch is small [1]. The output of the third stage is routed back to the input of the first (Fig. 5.1).

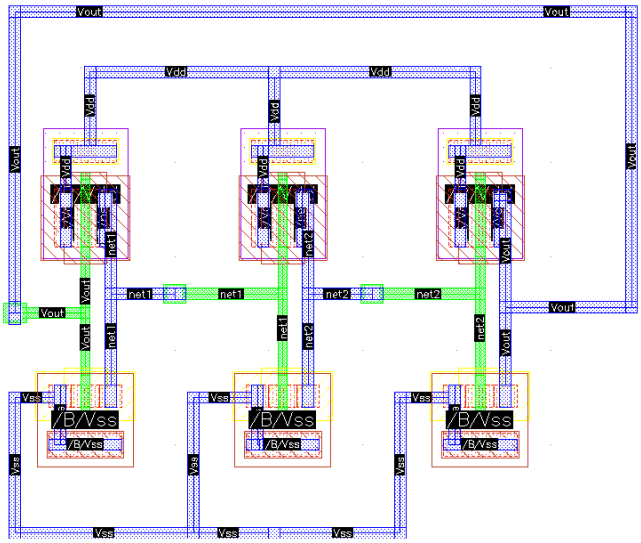


Figure 5.1: Layout of the 3-stage ring oscillator.

5.2 Physical Verification

Design Rule Check (DRC)

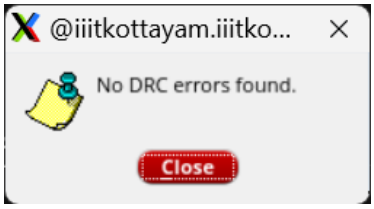


Figure 5.2: DRC result

DRC checks the layout against the design rules of the GPDK 90 nm process [1]. It reports no errors (Fig. 5.2), so the layout satisfies all the width, spacing and enclosure rules.

Layout versus Schematic (LVS)

LVS checks that the layout matches the schematic [1]. It reports "Schematic and Layout Match" with no net, device, pin or parameter mismatches (Fig. 5.3).

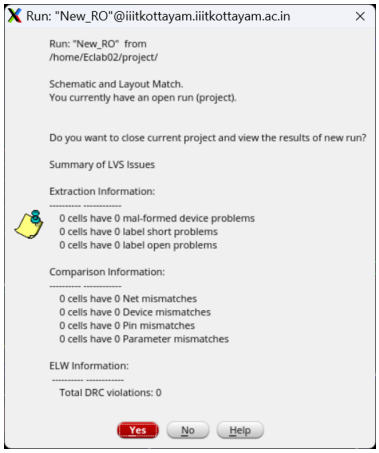


Figure 5.3: LVS result

5.3 Parasitic Extraction

Parasitic extraction obtains the resistance and capacitance of the wires and contacts [1].

The extraction run completes successfully (Fig. 5.4) and produces the `av_extracted` view, which is used for the post-layout simulation (Fig. 5.5).

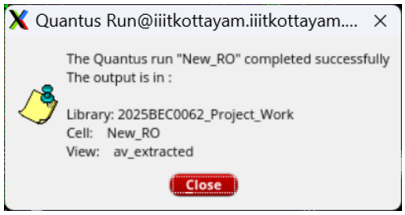


Figure 5.4: parasitic extraction run.

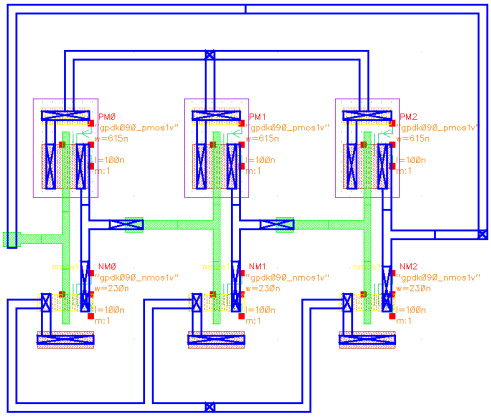


Figure 5.5: Extracted view of the ring oscillator.

6 Post-Layout Simulation

The transient simulation of Chapter 4 is repeated on the `av_extracted` view with the same supply voltage and temperature, so the wiring parasitics are now included [1].

6.1 Post-Layout Results

The transient simulation is run from 0 to 1 ns at 27°C, first at the nominal supply $V_{dd} = 1.2$ V (Fig. 6.1). The measured post-layout frequency is 10.87 GHz and the average power, taken from the supply current, is 180.12 μW.

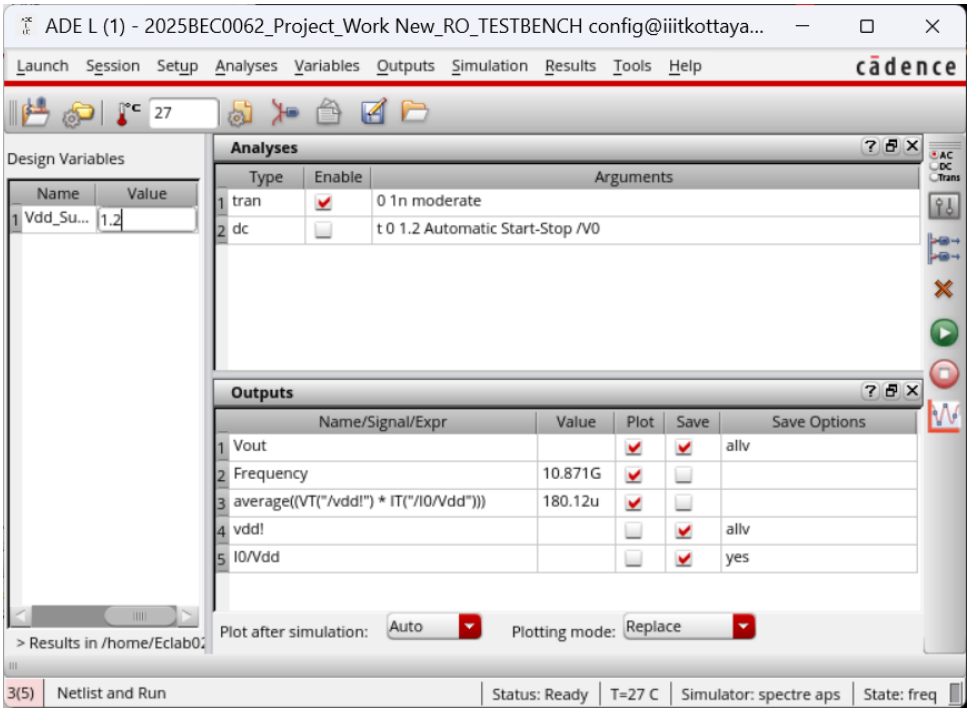


Figure 6.1: Post-layout simulation set-up and measured frequency at $V_{dd} = 1.2$ V.

Effect of Layout Parasitics In the schematic, each node has only the capacitance of the transistors. The layout adds the wiring capacitance C_{par}, so the frequency is multiplied by

$$\frac{f_{post}}{f_{pre}} = \frac{1}{1 + C_{par}/C} \tag{6.1}$$

which is always below one [2]. The frequency therefore falls after layout.

The output waveform is shown in Fig. 6.2.

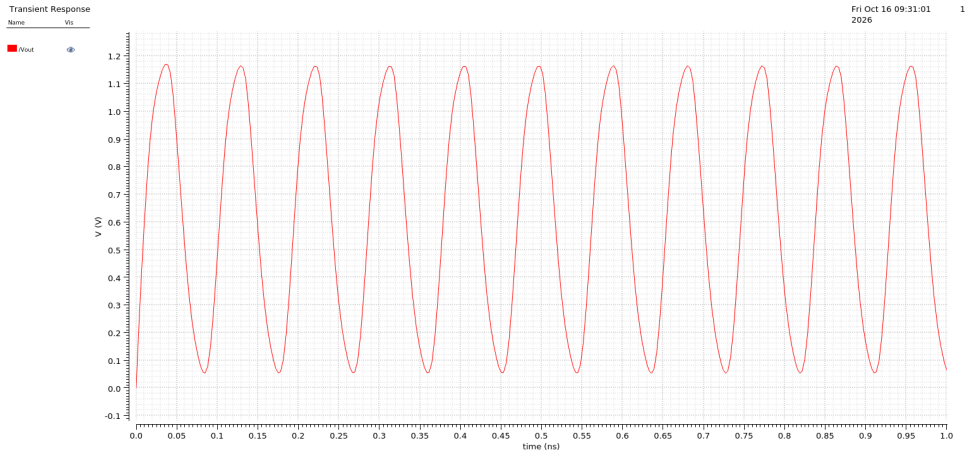


Figure 6.2: Post-layout transient response of the ring oscillator.

Re-tuning with the Supply Voltage : To check whether the lost frequency can be recovered, the post-layout simulation is repeated at $V_{dd} = 1.8$ V (Fig. 6.3). The frequency rises to 14.41 GHz, but the average power rises to 587.4 μW, about 3.3× the value at 1.2 V. The energy per cycle becomes 587.4 μW/14.41 GHz $\approx$ 40.8 fJ, compared with 16.6 fJ at 1.2 V.

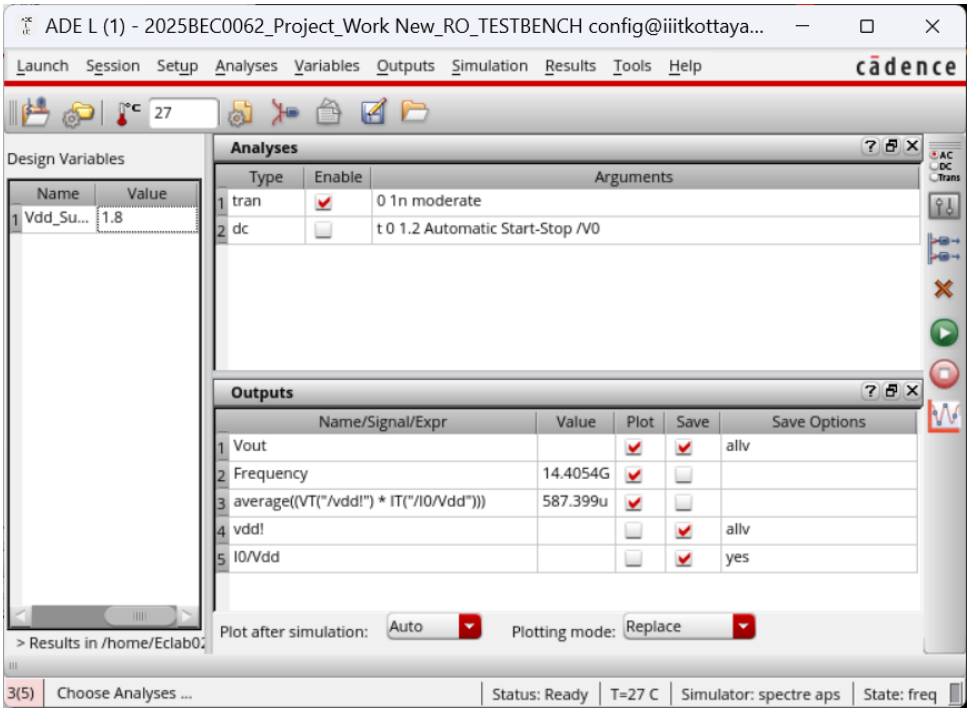


Figure 6.3: Post-layout simulation and measured frequency and power at $V_{dd} = 1.8$ V.

6.2 Pre-Layout VS Post-Layout

Table 6.1: Pre-layout and post-layout results.

Parameter	**Pre-layout** (1.2 V)	**Post-layout** (1.2 V)	**Post-layout** (1.8 V)*
Frequency (GHz)	14.72	10.87	14.41
Average power (μW)	182.5	180.12	587.4
Energy per cycle (fJ)	12.4	16.6	40.8

*Above the nominal supply; shown for trend only.

The post-layout frequency at 1.2 V is $10.87/14.72 \approx 0.74$ of the schematic value, a drop of about 26%. In [2] this ratio was about 0.7 for a minimum-size oscillator, so the result agrees with the expected effect of wiring capacitance. From Eq. (6.1), a ratio of 0.74 means $C_{par} \approx 0.35\,C$ (a first-order estimate).

The average power is almost unchanged after layout (182.5 μW to 180.12 μW) because the lower frequency offsets the extra capacitance ($P \propto CV_{dd}^2 f$). The energy per cycle therefore rises by about 34%, which reflects the added wiring capacitance. The pre-layout power was computed from the PMOS currents and the post-layout power from the supply current, so the two are approximately, not exactly, comparable.

Raising V_{dd} to 1.8 V recovers most of the lost frequency (14.41 GHz, about 96% of the target) but costs about 3.3× more power and exceeds the rated supply. The 15 GHz target is therefore not met after layout at any valid supply. Reaching it would need a layout with less wiring capacitance (shorter routes, tighter placement) or re-tuned device sizes.

7 PVT Analysis and Conclusion

The post-layout (`av_extracted`) oscillator was analysed across process, voltage and temperature (PVT). The process-corner and temperature analyses were run at the rated supply $V_{dd} = 1.2$ V, and the supply sweep covers 0.45 to 1.8 V.

Process (P): Corner Analysis. At $V_{dd} = 1.2$ V and 27°C, the oscillator starts and sustains oscillation in all five process corners (Fig. 7.1). The output stays within the 0 to 1.2 V rails in every corner. The frequency follows the order FF > NN > FS > SF > SS, from 7.75 GHz (SS) to 13.84 GHz (FF), which is −28.7% to +27.3% around NN (Table 7.1). The power changes in the same direction, so the energy per cycle varies only from 16.0 to 17.2 fJ. Even the fastest corner (FF) stays about 8% below the 15 GHz target, and the slowest corner (SS) is about 48% below it.

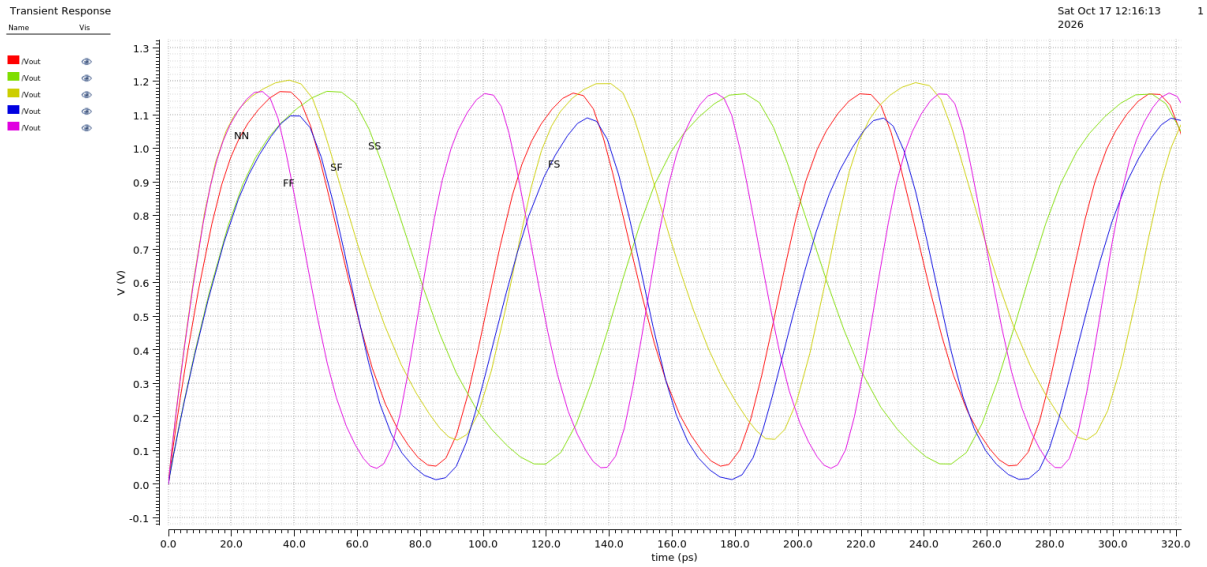


Figure 7.1: Post-layout transient response in five process corners ($V_{dd} = 1.2$ V, 27°C).

Table 7.1: Post-layout results in process corners ($V_{dd} = 1.2$ V, 27°C).

Corner	Freq. (GHz)	vs NN	Power (μW)	Energy (fJ)
FF	13.84	+27.3%	238.0	17.2
NN	10.87	–	180.1	16.6
FS	10.75	−1.1%	173.5	16.1
SF	10.01	−7.9%	163.5	16.3
SS	7.75	−28.7%	124.1	16.0

Voltage (V): Supply Analysis. The supply was swept from 0.45 to 1.8 V at 27°C in the NN corner (Fig. 7.2). The oscillator keeps running down to 0.45 V, where the frequency is about 1.6 GHz, and the frequency rises to about 7.8 GHz at 0.9 V, 12.0 GHz at 1.35 V and 14.4 GHz at 1.8 V. The slope is about 9.3 GHz/V between 0.9 and 1.35 V and falls to about 5.3 GHz/V above 1.35 V, because the gate overdrive grows more slowly at higher supply. Near the rated 1.2 V supply, a ±10% supply change (±0.12 V) therefore shifts the frequency by about ±1.1 GHz, or roughly ±10%.

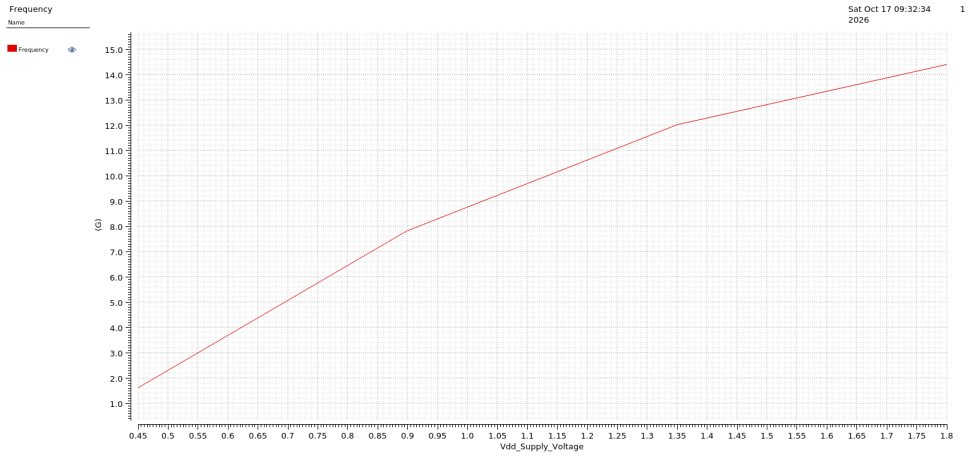


Figure 7.2: Post-layout oscillation frequency versus supply voltage (27°C, NN corner).

Temperature (T): Analysis. At $V_{dd} = 1.2$ V in the NN corner, the frequency falls almost linearly from about 11.9 GHz at 0°C to 10.87 GHz at 27°C and about 8.7 GHz at 100°C (Fig. 7.3). This is a drop of about 27% over the range, or roughly 32 MHz/°C, because carrier mobility decreases at high temperature and reduces the drive current of both transistors.

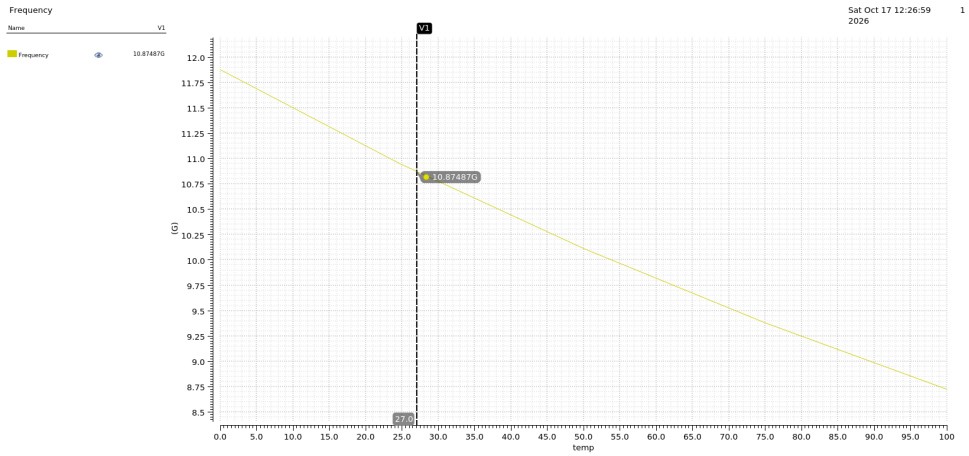


Figure 7.3: Post-layout frequency versus temperature ($V_{dd} = 1.2$ V, NN corner).

A 3-stage CMOS ring oscillator was designed in GPDK 90 nm through the complete

full-custom flow, with $W_n = 230$ nm, $W_p = 615$ nm, $L = 100$ nm and $V_{dd} = 1.2$ V. The schematic oscillates at 14.72 GHz (within about 2% of the 15 GHz target) with 182.5 μW and 12.4 fJ per cycle.

The layout passes DRC and LVS. After extraction the frequency falls to 10.87 GHz (-26%) because of wiring capacitance, with 180.12 μW and 16.6 fJ per cycle. Raising the supply to 1.8 V (trend only) recovers 14.41 GHz, about 96% of the target, at about 3.3$\times$ the power.

The PVT analysis shows that the post-layout oscillator starts in all five process corners at the rated supply, with the frequency ranging from 7.75 GHz (SS) to 13.84 GHz (FF), keeps running down to a 0.45 V supply, and loses about 27% of its frequency from 0 to 100°C. Layout effects should be considered early, and shorter routes, tighter placement or re-tuned sizes can recover the lost frequency within the rated supply.

References

[1] B. C. Prakruthi and M. V. Mandi, "Design and implementation of ring oscillator using Cadence Virtuoso," in *2025 IEEE 6th Global Conference for Advancement in Technology (GCAT)*, Bangalore, India, Oct. 2025, doi: 10.1109/GCAT66372.2025.11368527.

[2] E. Zafarkhah, M. Maymandi-Nejad, and M. Zare, "Single-ended ring oscillators: analysis and design," *IET Circuits, Devices & Systems*, vol. 14, no. 6, pp. 869–875, 2020, doi: 10.1049/iet-cds.2019.0190.